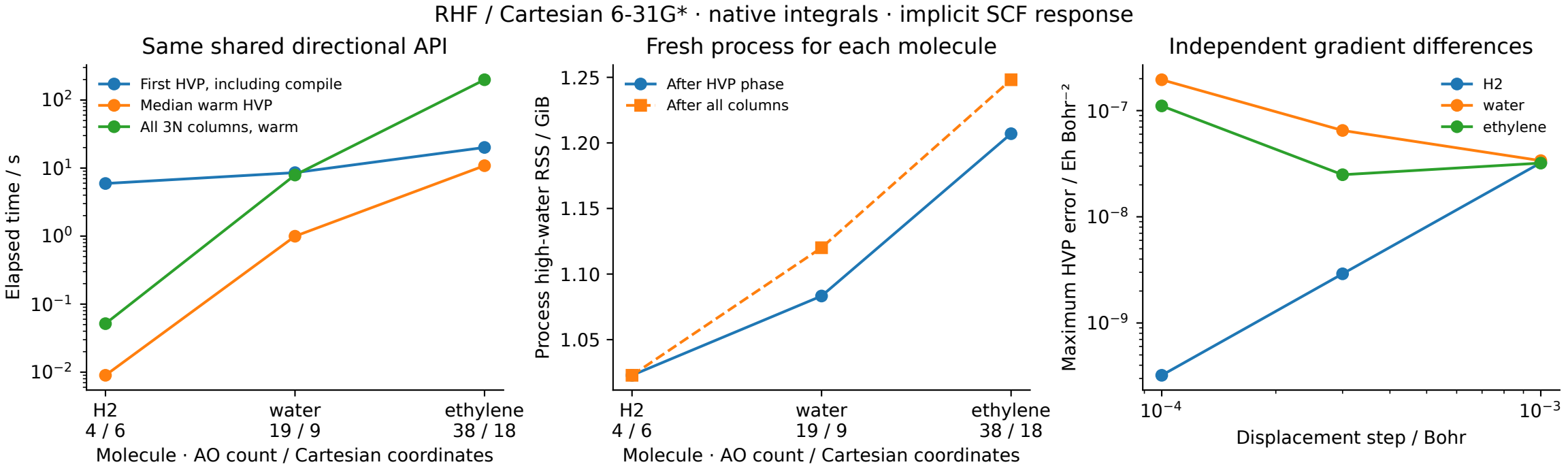
RHF / Cartesian 6-31G* · native integrals · implicit SCF response
Same shared directional API
First HVP, including compile
Median warm HVP
All 3N columns, warm
Elapsed time / s
$10^{2}$
$10^{1}$
$10^{0}$
$10^{-1}$
$10^{-2}$
H2
4 / 6
water
19 / 9
ethylene
38 / 18
Molecule · AO count / Cartesian coordinates
Fresh process for each molecule
After HVP phase
After all columns
Process high-water RSS / GiB
1.25
1.20
1.15
1.10
1.05
H2
4 / 6
water
19 / 9
ethylene
38 / 18
Molecule · AO count / Cartesian coordinates
Independent gradient differences
H2
water
ethylene
Maximum HVP error / Eh Bohr$^{-2}$
$10^{-7}$
$10^{-8}$
$10^{-9}$
$10^{-4}$
$10^{-3}$
Displacement step / Bohr